

11.1 Characteristics and difference between FIR and IIR system:

FIR System	IIR System
1. Finite duration impulse	1. Infinite duration impulse response
2. $h[n]=0$ for $(n<0$ and $n\geq M-1)$ then convolution output becomes $y[n] = \sum_{k=0}^{M-1} h[k]x[n-k] \text{ for FIR system.}$	2. $h[n]=0$ for $n<0$ so that convolution output becomes $y[n] = \sum_{k=0}^{\infty} h[k]x[n-k]$
3. FIR system has finite memory length.	3. IIR system has infinite memory length.
4. FIR system can be realized with finite number of addition, multiplication and memory location.	4. IIR system can't be realized with convolution sum because it requires infinite number of addition, multiplication and memory length. It is conveniently described by difference equation
5. Moving average system; $y[n] = \frac{1}{M} \sum_{k=0}^M x[n-k]$ is an example of FIR system.	5. Cumulative average given by, $y[n] = \frac{1}{n+1} \sum_{k=0}^n x[k]$ is an example of IIR system.

11.4 Direct Form realization of FIR filter:

Consider a causal FIR system,

$$y[n] = \sum_{k=0}^{M-1} h[k]x[n-k]$$

$$= h[0]x[n] + h[1]x[n-1] + h[2]x[n-2] + \dots + h[M-1]x[n-M+1]$$

Then the Direct Form realization of a causal FIR system is,

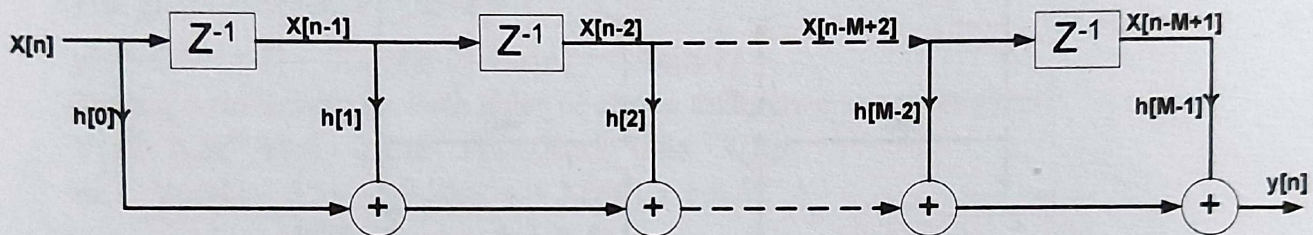


Figure 11.6: Direct Form realization of FIR system

Direct Form structures for IIR filters:

The output of a finite order LTI system at time n may be expressed as a linear combination of inputs and outputs. i.e.

$$y[n] = - \sum_{k=1}^N a_k y[n-k] + \sum_{k=0}^M b_k x[n-k]$$

Taking z-transform on both sides, we get,

$$Y(z) = - \sum_{k=1}^N a_k z^{-k} Y(z) + \sum_{k=0}^M b_k z^{-k} X(z)$$

$$\text{or, } Y(z) \left\{ 1 + \sum_{k=1}^N a_k z^{-k} \right\} = \sum_{k=0}^M b_k z^{-k} X(z)$$

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{\left\{ 1 + \sum_{k=1}^N a_k z^{-k} \right\}}$$

This transfer function can be represented in Direct Form I and II as below:

i. Direct Form I Structure:

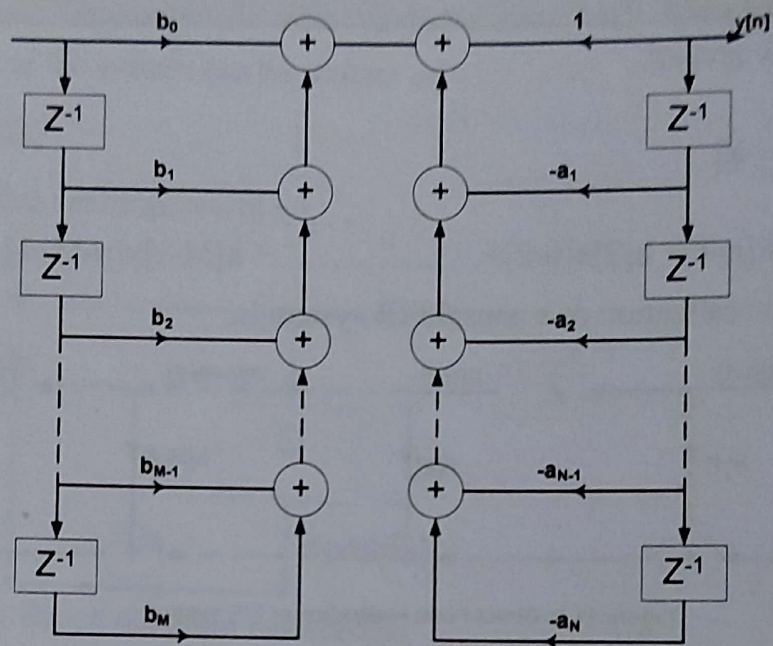


Figure 11.7 a): Direct form I structure for IIR filter

ii. Direct Form II structure:

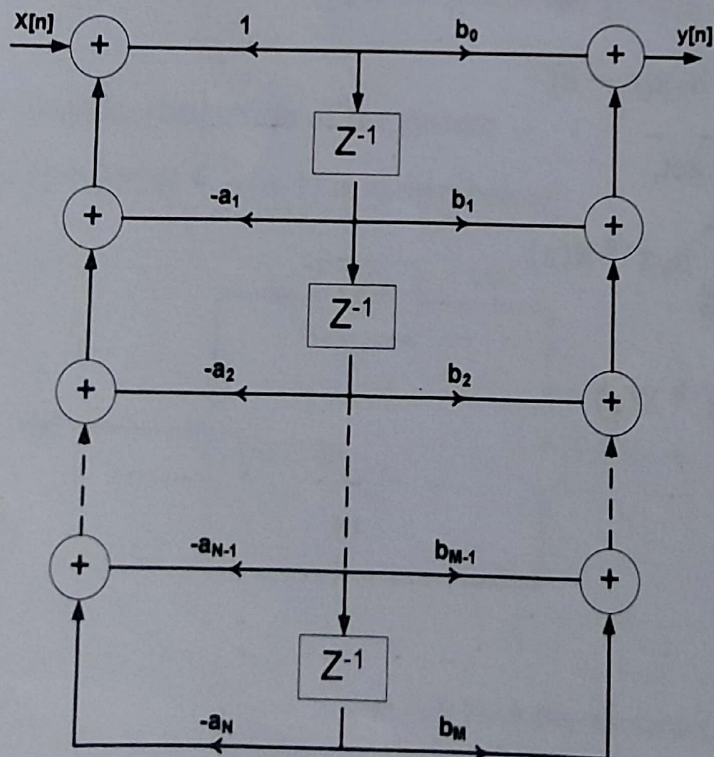


Figure 11.7 b): Direct form II structure for IIR filter

Above structures shows that Direct Form II realization requires less number of delays than Direct Form I. That means Direct Form II realization minimizes the number of memory locations than Direct Form I. Hence, Direct Form II is more efficient in terms of memory requirement than Direct Form I.

Example 11.6 Draw Direct Form I and Direct Form II of following difference equation:

i. $y[n] = 0.5y[n-1] - 0.25y[n-2] + x[n] + 0.4x[n-1]$

Solution:

The given difference equation is,

$$y[n] = 0.5y[n-1] - 0.25y[n-2] + x[n] + 0.4x[n-1]$$

Taking z-transform on both sides of above difference equation gives,

$$Y(z) = 0.5z^{-1}Y(z) - 0.25z^{-2}Y(z) + X(z) + 0.4z^{-1}X(z)$$

$$\text{or, } Y(z)\{1 - 0.5z^{-1} + 0.25z^{-2}\} = X(z)\{1 + 0.4z^{-1}\}$$

$$\therefore H(z) = \frac{Y(z)}{X(z)} = \frac{1 + 0.4z^{-1}}{1 - 0.5z^{-1} + 0.25z^{-2}}$$

Then its Direct Form I structure will be:

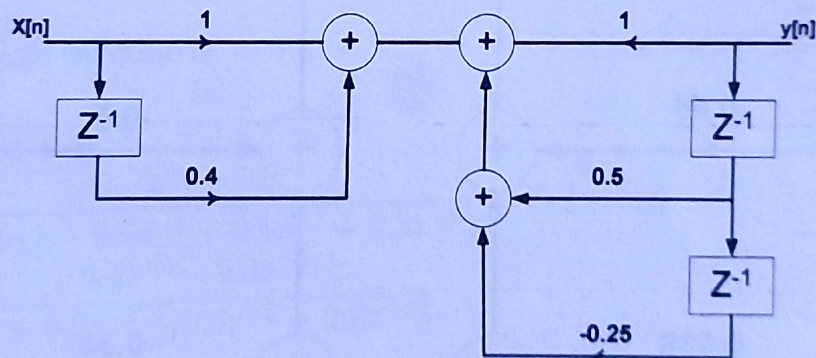


Figure 11.8 a): Direct form I structure

And Direct Form II will be:

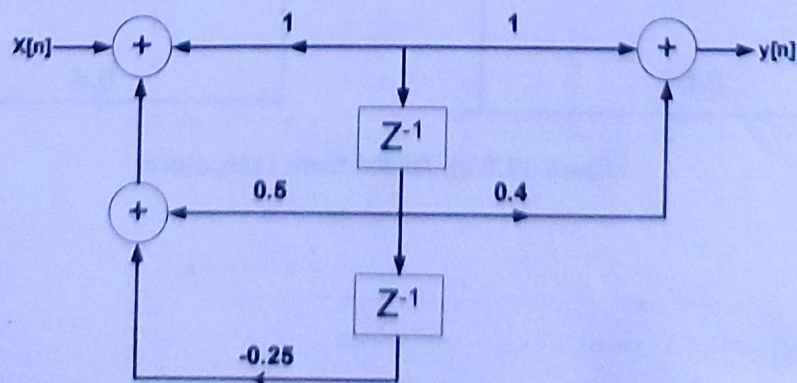


Figure 11.8 b): Direct form II structure

ii. $H(z) = \frac{0.14z^2 + 0.1595z + 0.02}{0.25z^3 + 0.15z^2 + 0.085z - 0.1}$

Solution:

The given transfer function is,

$$\begin{aligned} H(z) &= \frac{0.14z^2 + 0.1595z + 0.02}{0.25z^3 + 0.15z^2 + 0.085z - 0.1} \\ &= \frac{0.14z^2 + 0.1595z + 0.02}{0.25z^3 \{1 + 0.6z^{-1} + 0.34z^{-2} - 0.4z^{-3}\}} \\ &= \frac{0.56z^{-1} + 0.638z^{-2} + 0.08z^{-3}}{1 + 0.6z^{-1} + 0.34z^{-2} - 0.4z^{-3}} \end{aligned}$$

Then the Direct Form I structure is:

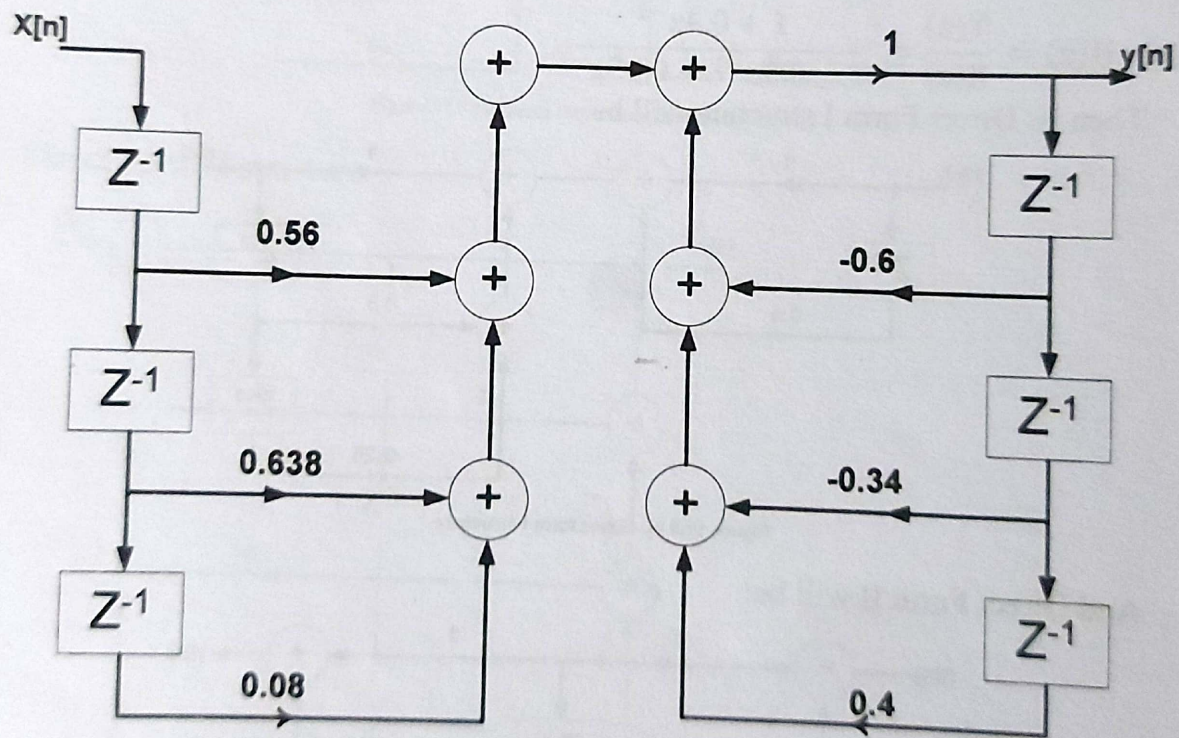


Figure 11.9 a): Direct form I structure

And Direct Form II structure is:

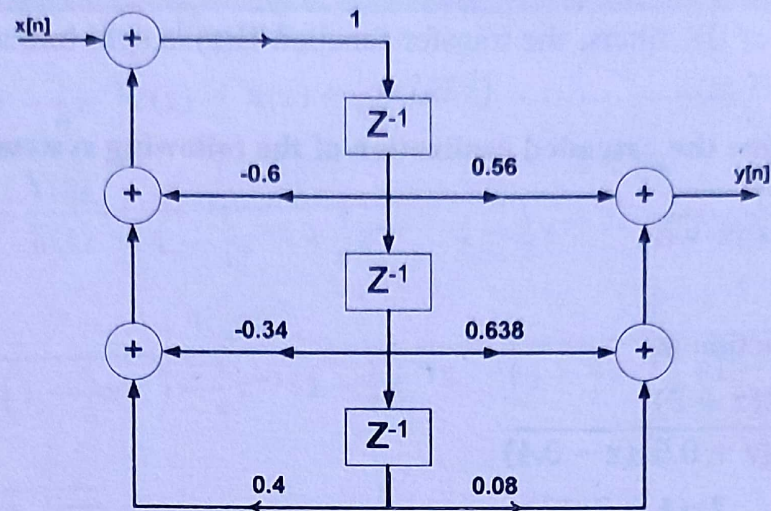


Figure 11.9 b): Direct form II structure

- iii. Draw Direct Form II structure of transfer function, $H(z) = \frac{z^{-1} - 3z^{-3}}{10 + 5z^{-1} + 5z^{-2} + 2z^{-3}}$.

Solution:

The given transfer function is,

$$H(z) = \frac{z^{-1} - 3z^{-3}}{10 + 5z^{-1} + 5z^{-2} + 2z^{-3}}$$

$$= \frac{z^{-1} - 3z^{-3}}{10\{1 + 0.5z^{-1} + 0.5z^{-2} + 0.2z^{-3}\}}$$

$$\therefore H(z) = \frac{0.1z^{-1} - 0.3z^{-3}}{\{1 + 0.5z^{-1} + 0.5z^{-2} + 0.2z^{-3}\}}$$

So, the Direct Form II structure is:

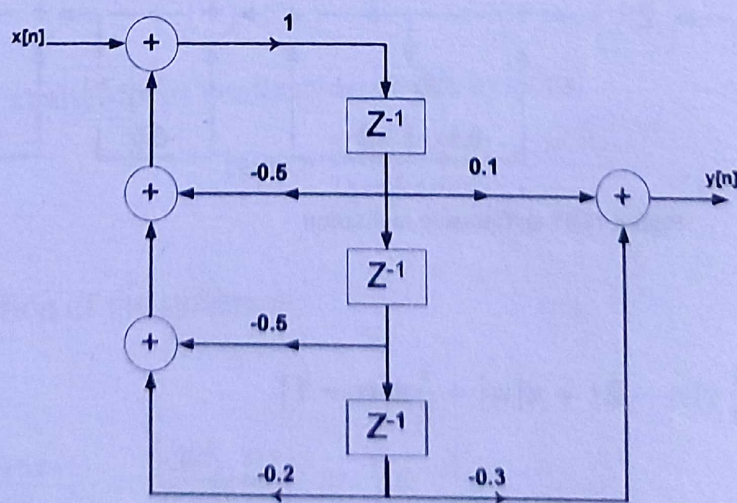


Figure 11.10 : Direct form II structure

11.5 Cascade Realization:

In cascade realization of IIR filters, the transfer function $H(z)$ is split into a product of transfer functions $H_1(z), H_2(z), H_3(z), \dots, H_k(z)$.

Example 9.7: Determine the cascaded realization of the following systems.

i. $H(z) = \frac{2(z+2)}{z(z-0.1)(z+0.5)(z-0.4)}$

Solution:

The given transfer function is,

$$\begin{aligned} H(z) &= \frac{2(z+2)}{z(z-0.1)(z+0.5)(z-0.4)} \\ &= \frac{2z(1+2z^{-1})}{z^4(1-0.1z^{-1})(1+0.5z^{-1})(1-0.4z^{-1})} \\ &= \frac{2z^{-3}(1+2z^{-1})}{(1-0.1z^{-1})(1+0.5z^{-1})(1-0.4z^{-1})} \\ &= \frac{2z^{-3}(1+2z^{-1})}{(1-0.1z^{-1})(1+0.5z^{-1})(1-0.4z^{-1})} \\ &= 2z^{-3} \cdot \frac{(1+2z^{-1})}{(1-0.1z^{-1})} \cdot \frac{1}{(1+0.5z^{-1})} \cdot \frac{1}{(1-0.4z^{-1})} \end{aligned}$$

Then its cascade realization is:

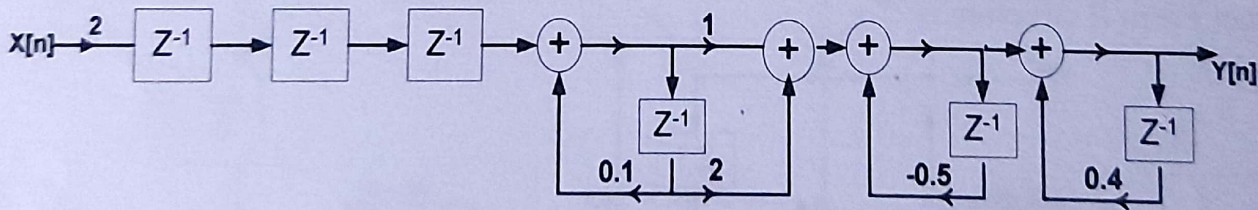


Figure 11.11 a): Cascade realization

ii. $y[n] = \frac{3}{4}y[n-1] - \frac{1}{8}y[n-2] + x[n] + \frac{1}{3}x[n-1]$

Solution:

The given difference equation is,

$$y[n] = \frac{3}{4}y[n-1] - \frac{1}{8}y[n-2] + x[n] + \frac{1}{3}x[n-1]$$

Taking z-transform on both sides, we get,

$$Y(z) = \frac{3}{4}z^{-1}Y(z) - \frac{1}{8}z^{-2}Y(z) + X(z) + \frac{1}{3}z^{-1}X(z)$$

$$\begin{aligned} \text{Or, } H(z) = \frac{Y(z)}{X(z)} &= \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}} = \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{1}{2}z^{-1} - \frac{1}{4}z^{-1} + \frac{1}{8}z^{-2}} \\ &= \frac{1 + \frac{1}{3}z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right) - \frac{1}{4}z^{-1}\left(1 - \frac{1}{2}z^{-1}\right)} = \frac{1 + \frac{1}{3}z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{4}z^{-1}\right)} \end{aligned}$$

$$\therefore H(z) = \frac{1 + \frac{1}{3}z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{1}{4}z^{-1}\right)} \cdot \frac{1}{\left(1 - \frac{1}{4}z^{-1}\right)}$$

Then its cascaded realization is:

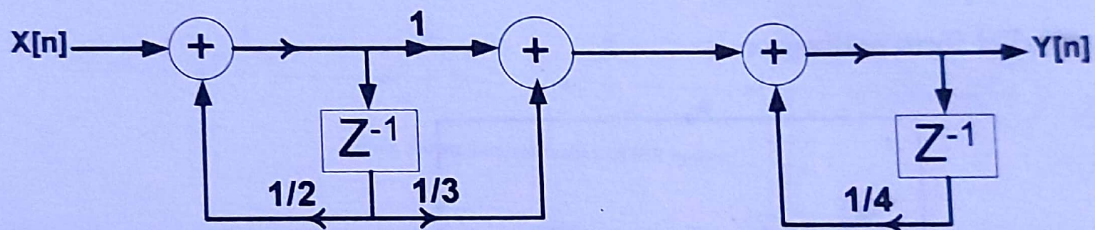


Figure 11.11 b): Cascade realization

11.6 Parallel Form Realization:

By the use of partial fraction expansion, the transfer function of an IIR filter can be realized in a parallel form.

Example 9.8: Obtain Parallel form realization of the system,

$$H(z) = \frac{3(2z^2 + 5z + 4)}{(2z + 1)(z + 2)}$$

Solution:

The given transfer function of the system is,

$$H(z) = \frac{3(2z^2 + 5z + 4)}{(2z + 1)(z + 2)}$$

$$\text{Let } F(z) = \frac{H(z)}{z} = \frac{3(2z^2 + 5z + 4)}{z(2z + 1)(z + 2)} = \frac{\frac{3}{2}(2z^2 + 5z + 4)}{\left(z + \frac{1}{2}\right)(z + 2)} = \frac{A}{z} + \frac{B}{z + \frac{1}{2}} + \frac{C}{z + 2}$$

Where,

$$A = z \cdot F(z)|_{z=0} = z \cdot \frac{\frac{3}{2}(2z^2+5z+4)}{(z+\frac{1}{2})(z+2)}|_{z=0} = 6$$

$$B = (z + \frac{1}{2}) \cdot F(z)|_{z=-\frac{1}{2}} = (z + \frac{1}{2}) \cdot \frac{\frac{3}{2}(2z^2+5z+4)}{(z+\frac{1}{2})(z+2)}|_{z=-\frac{1}{2}} = -4$$

$$C = (z + 2) \cdot F(z)|_{z=-2} = (z + 2) \cdot \frac{\frac{3}{2}(2z^2+5z+4)}{(z+\frac{1}{2})(z+2)}|_{z=-2} = 1$$

So,

$$F(z) = \frac{H(z)}{z} = \frac{6}{z} - \frac{4}{z+\frac{1}{2}} + \frac{1}{z+2}$$

Therefore,

$$H(z) = 6 - \frac{4z}{z+\frac{1}{2}} + \frac{z}{z+2} = 6 - \frac{4}{1+\frac{1}{2}z^{-1}} + \frac{1}{1+2z^{-1}}$$

So, its parallel form realization is:

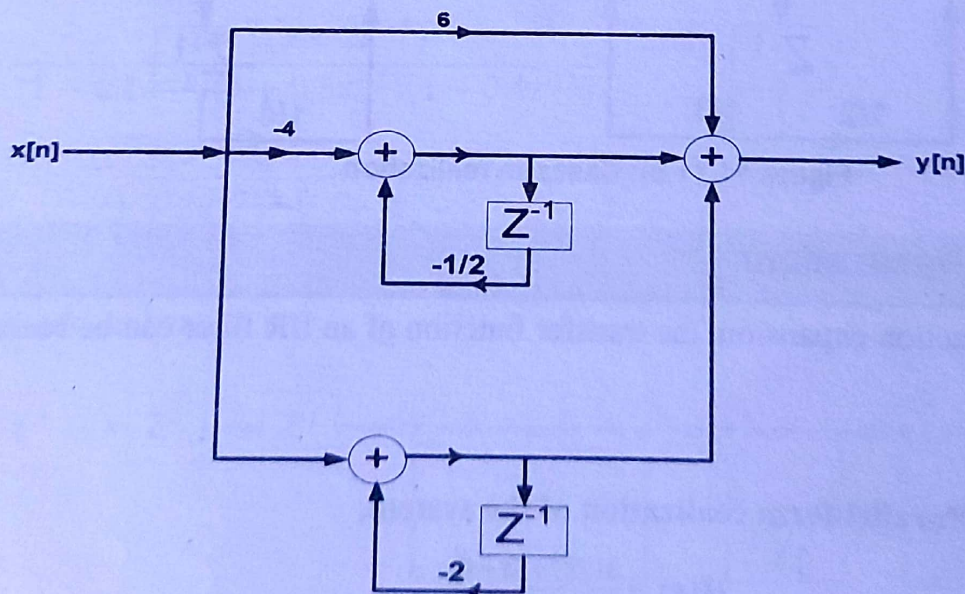


Figure 9.12: Parallel form realization

In general FIR system is described by the difference equation

$$y[n] = \sum_{k=0}^{M-1} b_k x[n-k]$$

Or, equivalently, the system function

$$H(z) = \sum_{k=0}^{M-1} b_k z^{-k}$$

Direct form structure

The DF realization follows convolution Summation,

$$y[n] = \sum_{k=0}^{M-1} h[k] x[n-k]$$

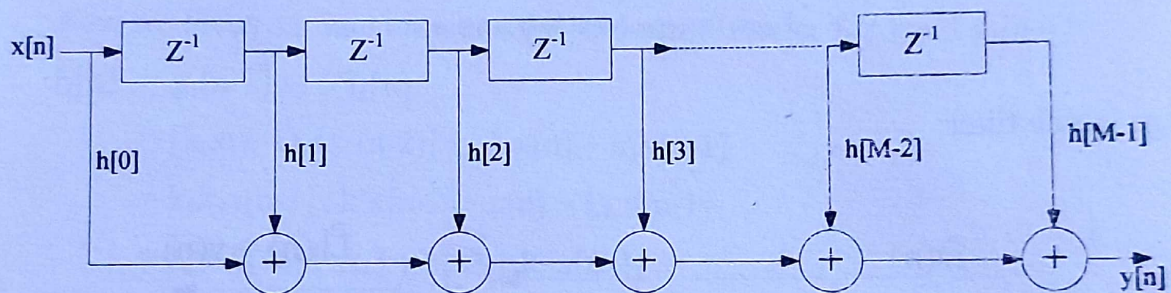


Figure: Direct form realization of FIR system

11.7 Lattice structure of an FIR Filter:

One of the FIR filter structure, called the lattice filter or lattice realization. Lattice filter are used extensively in digital speech processing and in the implementation of adaptive filters.

Let us begin the development by considering a sequence of FIR filters with system functions.

$$H_m(z) = A_m(z) \quad m = 0, 1, 2, \dots, M-1$$

Where by definition, $A_m(z)$ is the polynomial.

$$A_m(z) = 1 + \sum_{k=1}^m \alpha_m(k) z^{-k}$$

Where,

$$A_0(z) = 1, \alpha_m(k) \rightarrow \text{Filter coefficients}$$

If $\{x(n)\}$ is the input sequence to the filter $A_m(z)$ and $y[n]$ is the o/p of the sequence, we have,

$$y[n] = x[n] + \sum_{m=1}^m \alpha_m(k) x[n-k]$$

Direct form structure of FIR filter is illustrated as

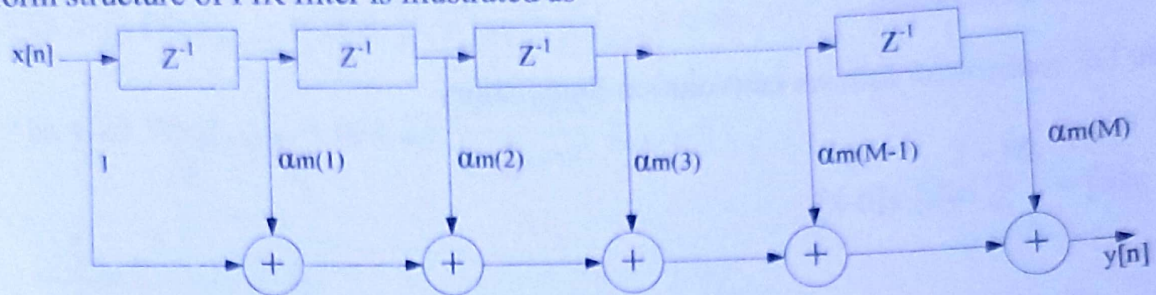


Figure: Direct form structure of FIR filter

Single stage lattice filter:

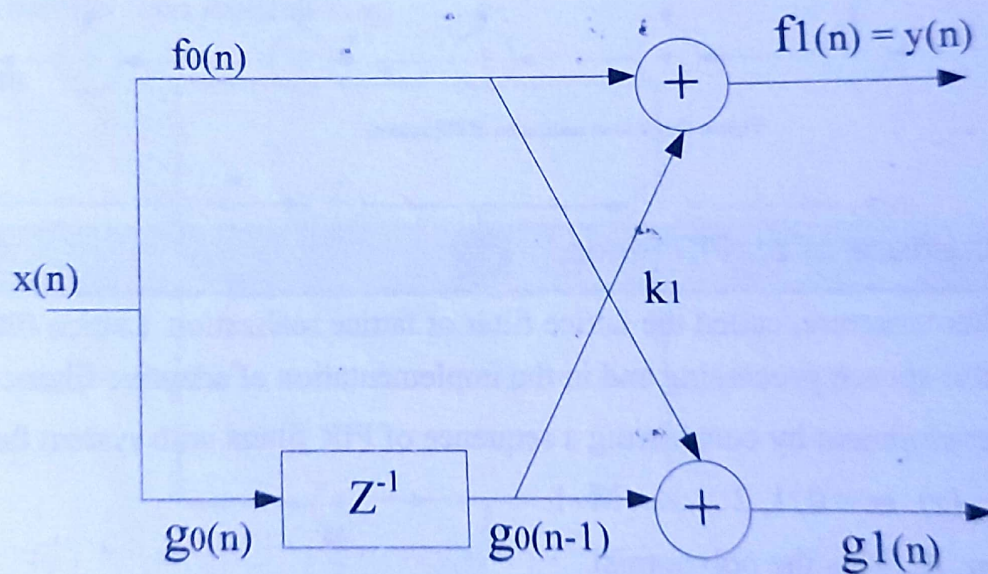


Figure: Single stage lattice structure

From figure,

$$f_0(n) = g_0(n) = x(n)$$

$$f_1[n] = f_0(n) + k_1 g[n-1] = x(n) + k_1 x[n-1]$$

$$g_1[n] = k_1 f_0(n) + g_0[n-1] = k_1 x[n] + x[n-1]$$

Two stage lattice filter:

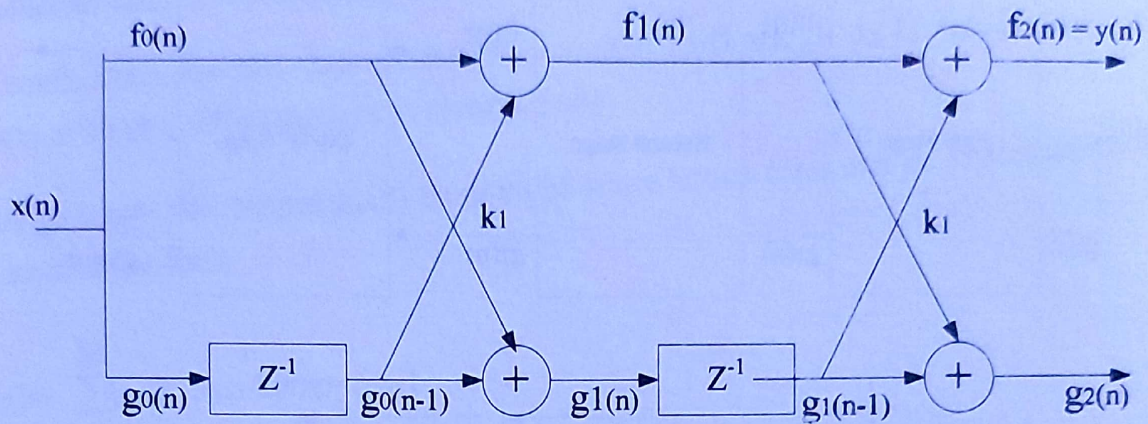


Figure: Two stage lattice structure

Now we focus our attention on $f_2[n]$ and substitute for $f_1[n]$ and $g_1[n-1]$

$$\begin{aligned}
 f_2[n] &= g_1[n-1] k_2 + f_1[n] \\
 &= [k_1 x(n-1) + x(n-2)] k_2 + x[n] + k_1 x[n-1] \\
 &= k_1 k_2 x[n-1] + k_2 x[n-2] + x[n] + k_1 x[n-1] \\
 &= x[n-1] k_1 [1+k_2] + x[n] + k_2 x[n-2] \\
 &= x[n] + k_1 [1+k_2] x[n-1] + k_2 x[n-2]
 \end{aligned}$$

So,

$$\alpha_2(2) = k_2, \alpha_2(1) = k_1(1+k_2)$$

or, equivalently,

$$k_2 = \alpha_2(2), k_1 = \frac{\alpha_2(1)}{1+\alpha_2(2)}$$

Where, k_1 and k_2 are referred to as reflection coefficients. Thus the lattice filter is generally described by the following set of equations,

$$f_0(n) = g_0(n) = x(n)$$

$$f_m(n) = f_{m-1}(n) + k_m g_{m-1}(n-1) \text{ where, } m = 1, 2, \dots, M-1$$

$$g_m(n) = k_m f_{m-1}(n) + g_{m-1}(n-1) \text{ where } m = 1, 2, \dots, M-1$$

Then the output of $(M-1)$ stage filter corresponds to the output of an $(M-1)$ order FIR filter, i.e.

$$y(n) = f_{m-1}(n)$$

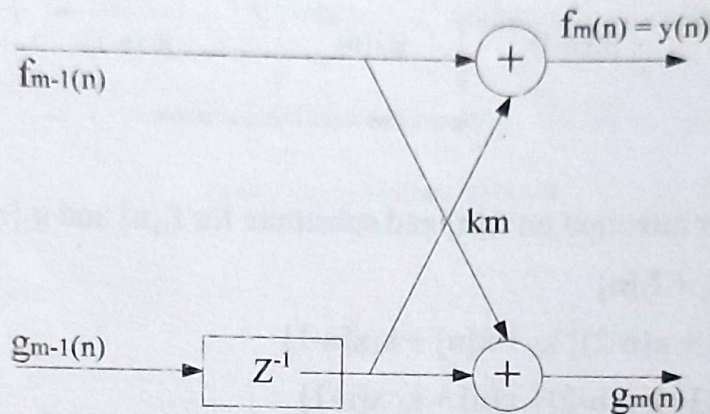
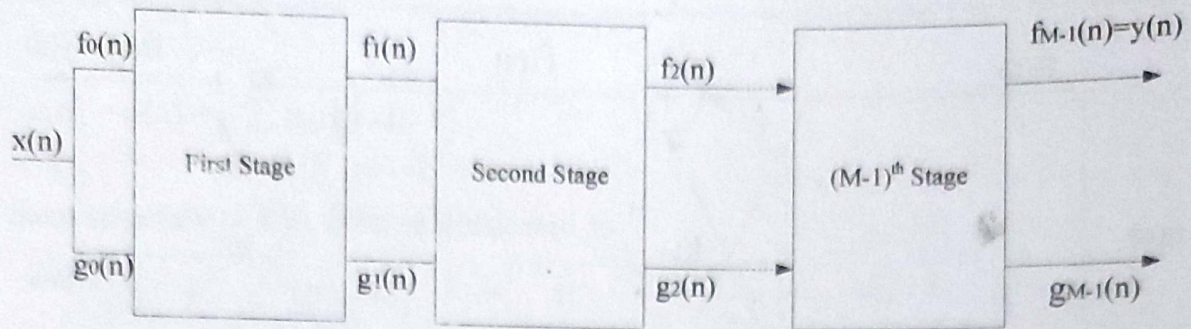


Figure: (M-1) stage lattice structure

As a consequence of the equivalence between an FIR filter and lattice filter, the output $f_m(n)$ of M- stage lattice filter can be expressed as.

$$f_m(n) = \sum_{k=0}^M \alpha_m(k) x[n-k], \quad \alpha_m(0) = 1$$

Since this equation is a convolution sum, the Z - transform follows.

$$F_m(z) = A_m(z) x(z)$$

$$\text{or, } A_m(z) = F_m(z) / x(z)$$

The other output component from the lattice namely, $g_m(n)$, can also be expressed in the form of a convolution sum by using another set of coefficients say $\beta_m(k)$

We know,

$$\begin{aligned} g_2[n] &= K_2 f_1[n] + g_1(n-1) \\ &= k_2 [x[n] + k_1 x[n-1]] + k_1 x[n-1] + x[n-2] \\ &= k_2 x[n] + k_1(1+k_2) x[n-1] + x[n-2] \\ &= a_2[2] x[n] + a_2[1] x[n-1] + x[n-2] \end{aligned}$$

Consequently, the filter coefficients are $\{a_2(2), a_2(1), 1\}$

Whereas the coefficients for filter that produces the output $f_2(n)$ are $\{1, a_2[1], a_2[2]\}$ where we see the two sets of filter coefficients are in reverse order.

Now we came to know that output $g_m(n)$ from an M-stage lattice filter can be expressed by the convolution sum of the form.

$$g_m(n) = \sum_{k=0}^M \beta_m(k) x(n-k)$$

Where the filter coefficients $\{\beta_m(k)\}$ are associated with filter that produces $f_m(n) = y(n)$ but operates in reverse order.

So,

$$\beta_m(k) = \alpha_m(M-k) \quad k = 0, 1, \dots, m, \beta_m(m) = 1$$

$$\therefore G_m(z) = B_m(z) x(z)$$

$$\text{Or, } B_m(z) = G_m(z) / x(z)$$

$$\text{Since, } \beta_m(k) = \alpha_m(M-k)$$

$$\therefore B_m(z) = \sum_{k=0}^M \alpha_m(M-k) z^{-k}$$

Let us assume $l = M-k$

$$B_m(z) = \sum_{l=0}^M \alpha_m(l) z^{l-M}$$

$$= z^{-M} \sum_{l=0}^M \alpha_m(l) z^l = z^{-M} A_m(z^{-1})$$

so, we have,

$$F_0(z) = G_0(z) = X(z)$$

$$F_m(z) = F_{m-1}(z) + k_m z^{-1} G_{m-1}(z) \quad m=1, 2 \dots M-1$$

$$G_m(z) = k_m F_{m-1}(z) + z^{-1} G_{m-1}(z) \quad m=1, 2, \dots M-1$$

Dividing each equation by $X(z)$, we obtain

$$A_0(z) = B_0(z) = 1$$

$$A_m(z) = A_{m-1}(z) + K_m z^{-1} B_{m-1}(z)$$

$$B_m(z) = K_m A_{m-1}(z) + z^{-1} B_{m-1}(z)$$

Thus a lattice stage is described in the Z- domain by the matrix equation

$$\begin{bmatrix} A_m(Z) \\ B_m(Z) \end{bmatrix} = \begin{bmatrix} 1 & k_m \\ k_m & 1 \end{bmatrix} \begin{bmatrix} A_{m-1}(Z) \\ Z^{-1} B_{m-1}(Z) \end{bmatrix}$$

Hence, for the conversion of lattice coefficients to direct form filter coefficients.

$$A_0(z) = B_0(z) = 1$$

$$A_m(z) = A_{m-1}(z) + k_m z^{-1} B_{m-1}(z) \quad m=1,2, \dots M-1$$

$$B_m(z) = z^{-m} A_m(z^{-1}) \quad m=1,2, \dots M-1$$

Let us take an example and clear the concept of; converting lattice coefficients to direct form filter coefficients and vice versa.

Note: For Stability the all the values of K_m should be less than 1

Example 11.9: Draw the lattice structure from the given FIR filter's system function. Also Check whether the stability of the system

$$H(z) = 1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3}$$

Solution,

Essential Formula

$$A_{m-1}(Z) = \frac{A_m(z) - K_m B_m(Z)}{1 - K_m^2} \dots\dots\dots (i)$$

$$B_m(z) = z^{-m} A_m(z^{-1}) \dots\dots\dots (ii)$$

Given,

$$H(z) = A_3(z) = 1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3}$$

$$\alpha_3(0) = 1,$$

$$\alpha_3(1) = \frac{13}{24}, \alpha_3(2) = \frac{5}{8}, \alpha_3(3)$$

$$= K_3 = 1/3$$

$$= \frac{4}{3} \left[1 + \frac{3}{8} Z^{-1} + \frac{1}{2} Z^{-2} - \frac{1}{4} - \frac{3}{16} Z^{-1} - \frac{1}{2} Z^{-2} \right]$$

$$= \frac{4}{3} \left[\frac{3}{4} + \frac{3}{16} Z^{-1} \right]$$

$$A_1(Z) = 1 + \frac{1}{4} Z^{-1}$$

$$\alpha_1(0) = 1, \alpha_1(1) = K_1 = \frac{1}{4}$$

Hence,

$$K_1 = \frac{1}{4}$$

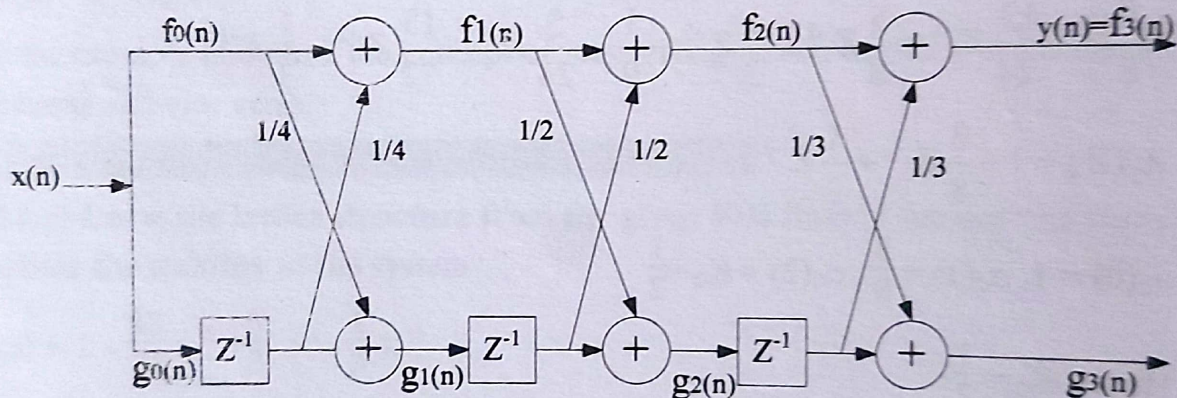


Figure: Lattice structure for all zero polynomial

In this example we converted direct form structure to lattice coefficients.

Since the value of k_1 , k_2 and k_3 are less than 1 the system is stable

Example 11.10: Given a 3 stage lattice filter with coefficients $K_1 = \frac{1}{4}$, $K_2 = \frac{1}{2}$ and $k_3 = \frac{1}{3}$.

Draw the corresponding FIR filter direct form I or II model.

Solution:

We know from recursive formula,

$$A_0(z) = B_0(z) = 1$$

$$A_m(z) = A_{m-1}(z) + k_m z^{-1} B_{m-1}(z)$$

and

$$B_m(z) = z^{-m} A_m(z^{-1})$$

Where,

$m = 1, 2, \dots, M-1$

So, For $m = 1$

$$A_1(z) = A_0(z) + k_1 z^{-1} B_0(z)$$

$$A_1(z) = 1 + \frac{1}{4} z^{-1}$$

Similarly,

$$B_1(z) = z^{-1} \left(1 + \frac{1}{4} z \right) = z^{-1} + \frac{1}{4}$$

$$A_2(z) = A_1(z) + K_2 z^{-1} B_1(z)$$

$$\text{or, } A_2(z) = 1 + \frac{1}{4} z^{-1} + \frac{1}{2} z^{-1} \left[\frac{1}{4} + z^{-1} \right]$$

$$= 1 + \frac{1}{4} z^{-1} + \frac{1}{8} z^{-1} + \frac{1}{2} z^{-2}$$

$$A_2(z) = 1 + \frac{3}{8} z^{-1} + \frac{1}{2} z^{-2}$$

And,

$$B_2(z) = z^{-2} \left[1 + \frac{3}{8} z + \frac{1}{2} z^2 \right]$$

$$B_2(z) = \frac{1}{2} + \frac{3}{8} z^{-1} + z^{-2}$$

$A_3(z)$

$$\text{or, } A_3(z) = 1 + \frac{3}{8} z^{-1} + \frac{1}{2} z^{-2} + \frac{1}{3} z^{-1} \left[\frac{1}{2} + \frac{3}{8} z^{-1} + z^{-2} \right]$$

$$\text{or, } A_3(z) = 1 + \frac{3}{8} z^{-1} + \frac{1}{2} z^{-2} + \frac{1}{6} z^{-2} + \frac{1}{3} z^{-3}$$

$$H(z) = A_3(z) = 1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3}$$

Where,

$m = 1, 2, \dots, M-1$

So, For $m = 1$

$$A_1(z) = A_0(z) + k_1 z^{-1} B_0(z)$$

$$A_1(z) = 1 + \frac{1}{4} z^{-1}$$

Similarly,

$$B_1(z) = z^{-1} \left(1 + \frac{1}{4} z \right) = z^{-1} + \frac{1}{4}$$

$$A_2(z) = A_1(z) + K_2 z^{-1} B_1(z)$$

$$\text{or, } A_2(z) = 1 + \frac{1}{4} z^{-1} + \frac{1}{2} z^{-1} \left[\frac{1}{4} + z^{-1} \right]$$

$$= 1 + \frac{1}{4} z^{-1} + \frac{1}{8} z^{-1} + \frac{1}{2} z^{-2}$$

$$A_2(z) = 1 + \frac{3}{8} z^{-1} + \frac{1}{2} z^{-2}$$

And,

$$B_2(z) = z^{-2} \left[1 + \frac{3}{8} z + \frac{1}{2} z^2 \right]$$

$$B_2(z) = \frac{1}{2} + \frac{3}{8} z^{-1} + z^{-2}$$

$A_3(z)$

$$\text{or, } A_3(z) = 1 + \frac{3}{8} z^{-1} + \frac{1}{2} z^{-2} + \frac{1}{3} z^{-1} \left[\frac{1}{2} + \frac{3}{8} z^{-1} + z^{-2} \right]$$

$$\text{or, } A_3(z) = 1 + \frac{3}{8} z^{-1} + \frac{1}{2} z^{-2} + \frac{1}{6} z^{-2} + \frac{1}{3} z^{-3}$$

$$H(z) = A_3(z) = 1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3}$$

11.8 Lattice Structure for All Pole System

$$\text{Let } H(z) = \frac{1}{\sum_{k=0}^M \alpha_m(k) x[n-k]} = \frac{1}{A_m(z)}$$

$$\text{Or, } A_m(z) = \sum_{k=0}^M \alpha_m(k) x[n-k]$$

The Lattice coefficient for all pole polynomial $A_m(z)$ will be identical to Lattice (or reflection coefficients) of all zero polynomial but the structural diagram is different.

Example 11.11: Compute the Lattice coefficients and draw the lattice structure for the following system function

$$H(z) = \frac{1}{1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}}$$

$$\text{Solution: } H(z) = \frac{1}{A_m(z)}$$

$$A_3(z) = 1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3}$$

Essential Formula

$$A_{m-1}(z) = \frac{A_m(z) - K_m B_m(z)}{1 - K_m^2} \dots\dots\dots (i)$$

$$B_m(z) = z^{-m} A_m(z^{-1}) \dots\dots\dots (ii)$$

$$\alpha_3(0) = 1, \alpha_3(1) = \frac{13}{24}, \alpha_3(2) = \frac{5}{8}, \alpha_3(3) = K_3 = 1/3$$

$$B_3(z) = z^{-3} \left[1 + \frac{13}{24} z + \frac{5}{8} z^2 + \frac{1}{3} z^3 \right]$$

$$= \frac{1}{3} + \frac{5}{8} z^{-1} + \frac{13}{24} z^{-2} + z^{-3}$$

Now, For $M=3$ from equation (i)

$$A_2(z) = \frac{A_3(z) - K_3 B_3(z)}{1 - K_3^2}$$

$$= \frac{[1 + \frac{13}{24}Z^{-1} + \frac{5}{8}Z^{-2} + \frac{1}{3}Z^{-3}] - \frac{1}{3}[\frac{1}{3} + \frac{5}{8}Z^{-1} + \frac{13}{24}Z^{-2} + Z^{-3}]}{1 - \frac{1}{9}}$$

$$= \frac{9}{8}[1 + \frac{13}{24}Z^{-1} + \frac{5}{8}Z^{-2} + \frac{1}{3}Z^{-3} - \frac{1}{9}] = \frac{5}{24}Z^{-1} - \frac{13}{72}Z^{-2} - \frac{1}{3}Z^{-3}]$$

$$A_2(Z) = 1 + \frac{3}{8}Z^{-1} + \frac{1}{2}Z^{-2}$$

$$\alpha_2(0) = 1, \alpha_2(1) = \frac{3}{8}, \alpha_2(2) = K_2 = \frac{1}{2}$$

$$\text{Hence, } K_2 = \frac{1}{2}$$

Again,

$$B_2(z) = z^{-2}[1 + \frac{3}{8}Z + \frac{1}{2}Z^2]$$

$$B_2(z) = \frac{1}{2} + \frac{3}{8}z^{-1} + z^{-2}$$

$$A_1(z) = \frac{A_2(z) - K_2 B_2(z)}{1 - K_2^2}$$

$$= \frac{1 + \frac{3}{8}Z^{-1} + \frac{1}{2}Z^{-2} - \frac{1}{2}[\frac{1}{2} + \frac{3}{8}Z^{-1} + Z^{-2}]}{1 - \frac{1}{4}}$$

$$= \frac{4}{3}[1 + \frac{3}{8}Z^{-1} + \frac{1}{2}Z^{-2} - \frac{1}{4} - \frac{3}{16}Z^{-1} - \frac{1}{2}Z^{-2}]$$

$$= \frac{4}{3}[\frac{3}{4} + \frac{3}{16}Z^{-1}]$$

$$A_1(Z) = 1 + \frac{1}{4}Z^{-1}$$

$$\alpha_1(0) = 1, \alpha_1(1) = K_1 = \frac{1}{4}$$

Hence,

$$K_1 = \frac{1}{4}$$

Therefore the lattice coefficients are $k_1 = \frac{1}{4}$, $k_2 = \frac{1}{2}$ and $k_3 = \frac{1}{3}$

The lattice structure is shown below:

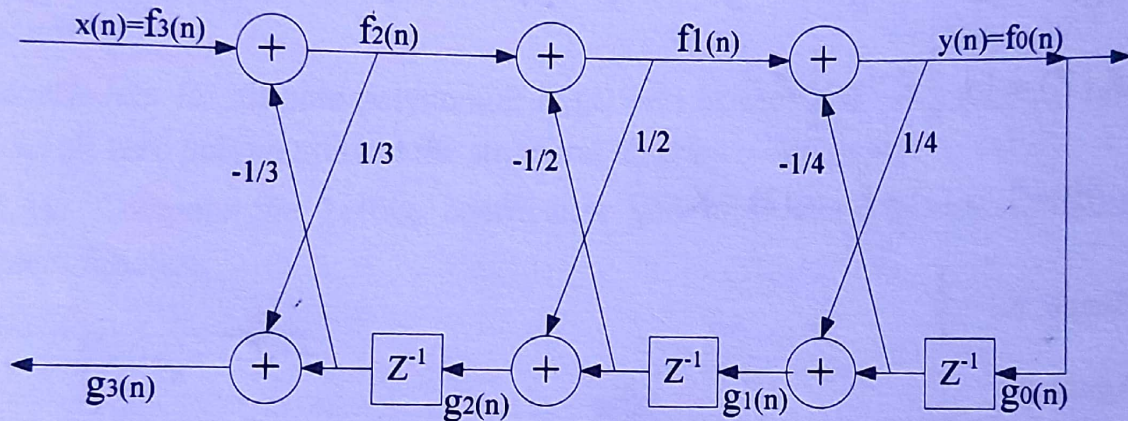


Figure: Lattice structure for all pole polynomial

Example 11.12: Compute reflection coefficients and draw the lattice structure for the system described as:

$$H(z) = 2 + \frac{36}{24}z^{-1} + \frac{10}{8}z^{-2} + \frac{2}{3}z^{-3}$$

Solution:

Since we have derived all the formula required to convert direct form structure in the form of $1 + Az^{-1} + Bz^{-2} + Cz^{-3} + \dots$ we need to convert the system function $H(z)$ in the similar form.

$$\text{Or, } H(z) = 2 \left[1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3} \right]$$

$$\text{Now we take } A_3(z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$$

Essential Formula

$$A_{m-1}(Z) = \frac{A_m(z) - K_m B_m(Z)}{1 - K_m^2} \dots \dots \dots (i)$$

$$B_m(z) = z^{-m} A_m(z^{-1}) \dots \dots \dots (ii)$$

$$\alpha_3(0) = 1, \alpha_3(1) = \frac{13}{24}, \alpha_3(2) = \frac{5}{8}, \alpha_3(3) = K_3 = 1/3$$

$$B_3(z) = z^{-3} \left[1 + \frac{13}{24} z + \frac{5}{8} z^2 + \frac{1}{3} z^3 \right]$$

$$= \frac{1}{3} + \frac{5}{8} z^{-1} + \frac{13}{24} z^{-2} + z^{-3}$$

Now, For M= 3 from equation (i)

$$A_2(z) = \frac{A_3(z) - K_3 B_3(z)}{1 - K_3^2}$$

$$= \frac{\left[1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3} \right] - \frac{1}{3} \left[\frac{1}{3} + \frac{5}{8} z^{-1} + \frac{13}{24} z^{-2} + z^{-3} \right]}{1 - \frac{1}{9}}$$

$$= \frac{9}{8} \left[1 + \frac{13}{24} z^{-1} + \frac{5}{8} z^{-2} + \frac{1}{3} z^{-3} \right] - \frac{1}{9} = \frac{5}{24} z^{-1} - \frac{13}{72} z^{-2} - \frac{1}{3} z^{-3}$$

$$A_2(z) = 1 + \frac{3}{8} z^{-1} + \frac{1}{2} z^{-2}$$

$$\alpha_2(0) = 1, \alpha_2(1) = \frac{3}{8}, \alpha_2(2) = K_2 = \frac{1}{2}$$

$$\text{Hence, } K_2 = \frac{1}{2}$$

Again,

$$B_2(z) = z^{-2} \left[1 + \frac{3}{8} z + \frac{1}{2} z^2 \right]$$

$$B_2(z) = \frac{1}{2} + \frac{3}{8} z^{-1} + z^{-2}$$

$$A_1(z) = \frac{A_2(z) - K_2 B_2(z)}{1 - K_2^2}$$

$$= \frac{1 + \frac{3}{8} z^{-1} + \frac{1}{2} z^{-2} - \frac{1}{2} \left[\frac{1}{2} + \frac{3}{8} z^{-1} + z^{-2} \right]}{1 - \frac{1}{4}}$$

$$= \frac{4}{3} \left[1 + \frac{3}{8} Z^{-1} + \frac{1}{2} Z^{-2} - \frac{1}{4} - \frac{3}{16} Z^{-1} - \frac{1}{2} Z^{-2} \right]$$

$$= \frac{4}{3} \left[\frac{3}{4} + \frac{3}{16} Z^{-1} \right]$$

$$A_1(Z) = 1 + \frac{1}{4} Z^{-1}$$

$$\alpha_1(0) = 1, \alpha_1(1) = K_1 = \frac{1}{4}$$

Hence,

$$K_1 = \frac{1}{4}$$

Therefore the lattice coefficients are $k_1 = \frac{1}{4}$, $k_2 = \frac{1}{2}$ and $k_3 = \frac{1}{3}$

Now we need to adjust the factor (2) that we have taken as common so as to adjust for ease of computation of lattice coefficients. This factor is just adjusted only in the structure as shown below:

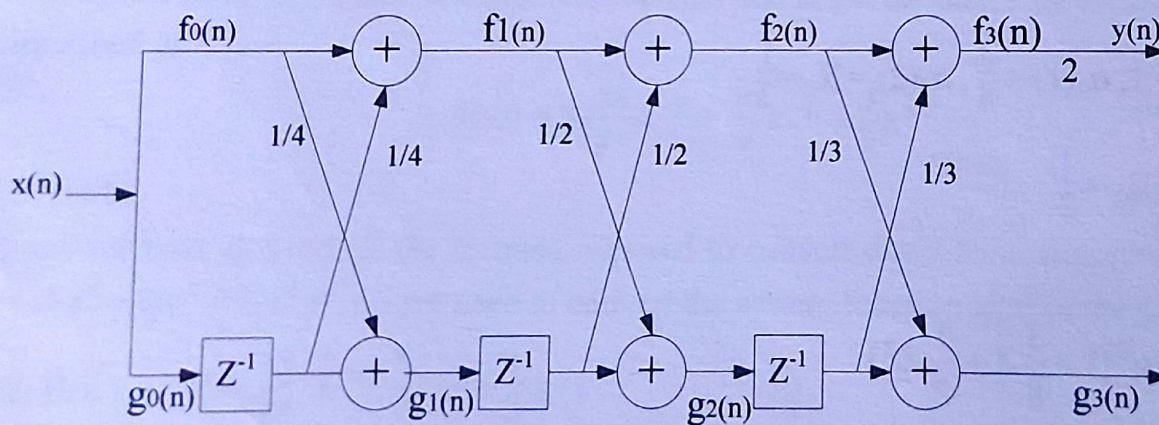


Figure: Lattice structure for all zero polynomial

LATTICE – LADDER STRUCTURE

Till now we have drawn the structure for either all pole or all zero polynomial but what for the pole – zero polynomial. For pole – zero polynomial we have a structure called Lattice – Ladder structure and here we begin with.

Let us consider a pole – zero polynomial $H(z) = \frac{B_m(z)}{A_m(z)}$.

For this type of system we follow the following algorithm:

1. Find the lattice coefficients for $\frac{1}{A_m(z)}$ all pole system and draw the structure
2. Sort out the coefficients of $B_m(z)$ i.e. $b_0, b_1, b_2, \dots, b_n$
3. Use $C_M = b_M = 1$, where M is the order of the polynomial for ladder
4. For other ladder coefficients use:

$$c_m = b_m - \sum_{i=m+1}^M c_i \alpha_i (i - m)$$

Where $m = M-1, M-2, \dots, 1, 0$

Example 11.13: Convert the following pole zero IIR filter into a lattice ladder structure.

$$H(z) = \frac{1+2z^{-1}+2z^{-2}+z^{-3}}{1+\frac{13}{24}z^{-1}+\frac{5}{8}z^{-2}+\frac{1}{3}z^{-3}}$$

Solution:

Let us first find out the lattice coefficients for all-pole polynomial.

$$\frac{1}{A_3(z)} = \frac{1}{1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}}$$

$$\text{So we have } A_3(z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$$

Essential Formula

$$A_{m-1}(Z) = \frac{A_m(z) - K_m B_m(Z)}{1 - K_m^2} \dots \dots \dots (i)$$

$$B_m(z) = z^{-m} A_m(z^{-1}) \dots \dots \dots (ii)$$

$$\alpha_3(0) = 1, \alpha_3(1) = \frac{13}{24}, \alpha_3(2) = \frac{5}{8}, \alpha_3(3) = K_3 = 1/3$$

$$B_3(z) = z^{-3} \left[1 + \frac{13}{24}z + \frac{5}{8}z^2 + \frac{1}{3}z^3 \right]$$

$$= \frac{1}{3} + \frac{5}{8}z^{-1} + \frac{13}{24}z^{-2} + z^{-3}$$

Now, For $M=3$ from equation $\dots \dots \dots (i)$

$$A_2(z) = \frac{A_3(z) - K_3 B_3(z)}{1 - K_3^2}$$

$$= \frac{[1 + \frac{13}{24}Z^{-1} + \frac{5}{8}Z^{-2} + \frac{1}{3}Z^{-3}] - \frac{1}{3}[\frac{1}{3} + \frac{5}{8}Z^{-1} + \frac{13}{24}Z^{-2} + Z^{-3}]}{1 - \frac{1}{9}}$$

$$= \frac{9}{8}[1 + \frac{13}{24}Z^{-1} + \frac{5}{8}Z^{-2} + \frac{1}{3}Z^{-3} - \frac{1}{9}] = \frac{5}{24}Z^{-1} - \frac{13}{72}Z^{-2} - \frac{1}{3}Z^{-3}]$$

$$A_2(Z) = 1 + \frac{3}{8}Z^{-1} + \frac{1}{2}Z^{-2}$$

$$\alpha_2(0) = 1, \alpha_2(1) = \frac{3}{8}, \alpha_2(2) = K_2 = \frac{1}{2}$$

$$\text{Hence, } K_2 = \frac{1}{2}$$

Again,

$$B_2(z) = z^{-2}[1 + \frac{3}{8}Z + \frac{1}{2}Z^2]$$

$$B_2(z) = \frac{1}{2} + \frac{3}{8}z^{-1} + z^{-2}$$

$$A_1(z) = \frac{A_2(z) - K_2 B_2(z)}{1 - K_2^2}$$

$$= \frac{1 + \frac{3}{8}Z^{-1} + \frac{1}{2}Z^{-2} - \frac{1}{2}[\frac{1}{2} + \frac{3}{8}Z^{-1} + Z^{-2}]}{1 - \frac{1}{4}}$$

$$= \frac{4}{3}[1 + \frac{3}{8}Z^{-1} + \frac{1}{2}Z^{-2} - \frac{1}{4} - \frac{3}{16}Z^{-1} - \frac{1}{2}Z^{-2}]$$

$$= \frac{4}{3}[\frac{3}{4} + \frac{3}{16}Z^{-1}]$$

$$A_1(Z) = 1 + \frac{1}{4}Z^{-1}$$

$$\alpha_1(0)=1, \alpha_1(1)=K_1=\frac{1}{4}$$

Hence,

$$K_1=\frac{1}{4}$$

Therefore the lattice coefficients are $k_1=\frac{1}{4}$, $k_2=\frac{1}{2}$ and $k_3=\frac{1}{3}$

We also have found out the following filter coefficients parameters:

$$\alpha_3(0)=1, \alpha_3(1)=\frac{13}{24}, \alpha_3(2)=\frac{5}{8}, \alpha_3(3)=\frac{1}{3}$$

$$\alpha_2(0)=1, \alpha_2(1)=\frac{3}{8}, \alpha_2(2)=\frac{1}{2}$$

$$\alpha_1(0)=1, \alpha_1(1)=\frac{1}{4}$$

Now for ladder the coefficients of all zero system i.e.

$$B_m(z)=1+2z^{-1}+2z^{-2}+z^{-3}$$

$$b_0=1, b_1=2, b_2=2, \text{ and } b_3=1$$

We know from the algorithm $c_3=b_3=1$

For other ladder coefficients, We have ,

$$c_m = b_m - \sum_{i=m+1}^M c_i \alpha_i(i-m)$$

For $m=2$

$$c_2 = b_2 - \sum_{i=3}^3 c_i \alpha_i(0)$$

$$c_2 = b_2 - c_3 \alpha_3(0) = \frac{35}{24}$$

Similarly, for $m=1$

$$c_1 = b_1 - \sum_{i=2}^3 c_i \alpha_i(i-1)$$

$$c_1 = b_1 - c_2 \alpha_2(1) - c_3 \alpha_3(2) = \frac{53}{64}, \text{ Similarly,}$$

$$C_0 = -0.2695$$

The lattice ladder structure is shown below:

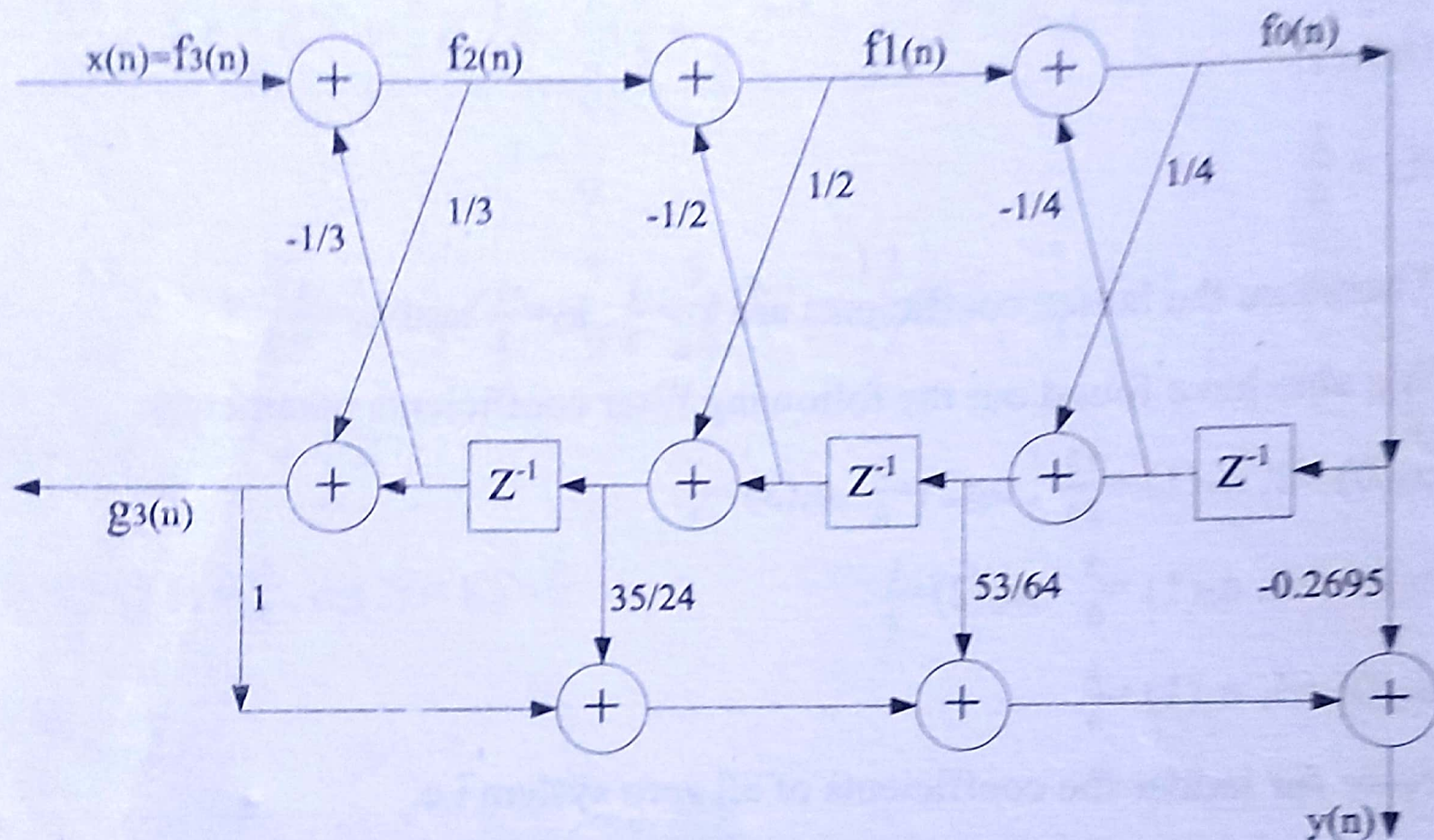


Figure: Lattice Ladder structure for pole – zero polynomial